

Sushan Adhikari

Computer Engineering Student | AI/ML | NLP | Computer Vision

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PROFILE

Final-year Computer Engineering student (B.Eng., Kathmandu University; exchange semester at IIT Palakkad) with research and industry experience spanning ML/MLOps, NLP, Computer Vision, LLMs, and RAG. First author on 3 publications (ICAIL 2026, KUSET); co-founded two AI-driven EdTech startups. Contributed to Nepal's national pension platform (300k+ users) and a global non-profit employment pipeline. Seeking PhD with RA/TA opportunities in AI/ML, NLP, LLMs, or Computer Vision.

EDUCATION

B.Eng. Computer Engineering — Kathmandu University, School of Engineering Jan 2022 – Feb 2026

CGPA: 3.88/4.0 (through 7th Semester) [\[KU Transcript\]](#)

Coursework: Data Structures & Algorithms, Computer Vision, Databases, Computer Architecture, Statistics & Probability, Differential Equations

Exchange Semester, B.Tech CSE — Indian Institute of Technology Palakkad

Jan – May 2025

Coursework: Introduction to AI, Linear Algebra, Compiler Design, Graph Theory & Combinatorics [\[IIT Transcript\]](#)

RESEARCH & PUBLICATIONS

Conference

Directional Asymmetry in Low-Resource Legal Machine Translation: A Nepali-English Case Study *ICAIL 2026, Apr 2026*

First Author · [\[GitHub\]](#) · [\[Acceptance Email\]](#)

- Curated the first publicly available Nepali-English legal parallel corpus (5,024 sentence pairs) and fine-tuned mBART-50 and NLLB-200 under unidirectional and bidirectional training regimes.
- Bidirectional training improves Nepali→English by ≈ 5 BLEU but degrades English→Nepali by 6.3 BLEU in mBART-50, attributing the asymmetry to Nepali's morphological complexity causing representational interference.

Journal

Enhancing Ethical Reasoning in Tiny LLMs via Fine-Tuning and Multi-Agent Consensus *KUSET Vol.18 No.2, Dec 2024*

First Author · [\[GitHub\]](#) · [\[NCCI Certificate\]](#)

- Fine-tuned TinyLlama-1.1B using LoRA on 1,000 synthetic ethical dilemmas; deployed three expert agents (utilitarian, deontological, virtue ethics) with a confidence-weighted consensus mechanism.
- Agents achieve 95+ % philosophical consistency scores (deontological: 97.8, utilitarian: 95.2, virtue ethics: 96.5), validated via human Likert-scale evaluation, demonstrating compact models can perform meaningful moral reasoning.

Detecting Image Forgeries and Deepfakes: A Comparative Study of CNN and Transformer Models *KUSET Vol.18 No.2, Dec 2024*

Co-Author (2nd)

- Benchmarked InceptionV3 and ViT on a custom 140,000+ image dataset covering photoshopping, splicing, copy-move, face-swapping, and StyleGAN/deepfake-generated content.
- InceptionV3 outperforms ViT by 5.51 % accuracy (94.0 % vs. 88.49 %) and 10.8 % recall, challenging the assumption that Transformers universally excel where local texture cues are discriminative.

Under Review / Ongoing

RAG for Theoretical CS Education: Algorithm Analysis & Complexity Theory

Under Review: USC 2025

First Author · [\[GitHub\]](#) · [\[USC Certificate\]](#)

- Built "AlgoRAG": a domain-specialized RAG system over 800+ lecture slides and 150+ proof templates targeting algorithm analysis and complexity theory with mathematical entity recognition.
- Achieved 100 % response success on 179 curated exam-style questions; pedagogical quality score 0.7620 (graph algorithms: 0.8086, NP-completeness: 0.7643); standard BLEU-4 scores zero due to proof expression variability.

Physics-Informed Data Augmentation for Sediment Concentration Prediction in Himalayan Rivers *Accepted: LEC / NepJOL*

3rd Author · [\[LEC Certificate\]](#)

- Compared 10 augmentation strategies (5 classical statistical, 5 physics-informed) for sediment concentration prediction in data-scarce Himalayan rivers critical to sustainable hydropower.
- Physics-informed methods outperform classical baselines: R^2 improved by 5.5 %, RMSE reduced by 24.3 %, MAE reduced by 47.6 %, validated through 5-fold cross-validation.

Automated Mathematical Animation Generation: Dataset Requirements and Validation Gaps *Ongoing (Submitted: Elsevier)*

First Author · [\[GitHub / Dataset\]](#)

- Fine-tuned DeepSeek-Coder-1.3B on a rigorously validated 537-sample Manim dataset; exposed a 100-percentage-point gap between weak (string matching) and rigorous (syntax + execution) validation.
- Training loss drops 0.536→0.067 but 0 % test-set success rate confirms a dataset bottleneck; 2,000–5,000 samples estimated for reliable generalization in a 1.3B-parameter model.

Explainability Methods for Hybrid CNN-Transformer Document Layout Analysis

Ongoing

First Author · [\[GitHub\]](#)

- Systematically compared Grad-CAM, Attention Rollout, and LIME for explaining hybrid CNN-Transformer document layout analysis models using PubLayNet and DocLayNet as benchmarks.
- CNN branch emphasizes local boundary and edge cues (headers, footers); Transformer captures global page organization (tables, columns); quantitative explainability metrics guide method selection by document type.

TECHNICAL SKILLS

Languages & ML: Python; PyTorch, Scikit-Learn, OpenCV, YOLOv8; Pandas, NumPy
Deployment/Inference: NVIDIA DeepStream, Triton Inference Server, Docker, AWS/EC2, Terraform, Git
Backend & Data: FastAPI, Flask, REST APIs, PostgreSQL, data preprocessing & annotation
Other: CI/CD basics, Spring Boot (integration), LaTeX/Overleaf

INTERNSHIP EXPERIENCE

- **Junior MLOps Developer** — KyraWorks *Jan – Apr 2026* [\[Certificate\]](#)
Built CI/CD pipelines using Docker, EC2, AWS, and Terraform for automated model deployments.
- **Computer Vision Intern** — KyraWorks *Oct 2025 – Jan 2026* [\[Certificate\]](#)
Deployed and optimized 5+ models on Triton Inference Server; benchmarked NVIDIA DeepStream and GStreamer for real-time multi-camera (10+) frame processing.
- **AI and Data Engineer** — [Mercuri.world](#) (*Non-profit, Volunteer*) *Jan 2025 – Present*
Built ML-based job recommendation pipeline for 50+ job types; improved employment access for individuals with mental illness; collaborated with 150+ global volunteers.
- **Full-Stack Development Intern** — National Pension Management System, Nepal *Jun – Oct 2025* [\[Certificate\]](#)
Contributed to national pension platform (300k+ users); implemented anti-spoofing liveness detection (Amazon Rekognition); assisted in DB schema design (Spring Boot, PostgreSQL).

PROJECTS & STARTUPS

- **Nurvexa** — Co-founder [nurvexalabs.com](#) *2025 – Present*
RAG-powered nursing exam preparation platform; adaptive question generation and personalized study plans for the PCL Nursing licensing exam.
- **PrepGraduate** — Co-founder [prepgraduate.com](#) *2026 – Present*
AI-driven GRE preparation platform with full-length mock tests, adaptive practice, and score analytics targeting graduate program applicants worldwide.

CERTIFICATIONS

[Machine Learning Specialization](#) — Andrew Ng (Stanford/Coursera, 2026) · [Supervised & Unsupervised Learning, AI Fundamentals, PySpark, SQL](#) (DataCamp, 2025) · [Embeddings for NLP](#) (OpenHPI, 2025)

ACHIEVEMENTS & LEADERSHIP

- **Hult Prize OnCampus Winner** (2024) — Led CV pipeline for fish-disease classification (10k+ images, ~60% baseline accuracy); represented Nepal at Regional Summit, Bangkok, Thailand.
- **Finalist** — ICT Awards 2024 (Rising Star Innovation Category).
- **People's Choice Winner** — NASA Space Apps Challenge 2024. [\[Certificate\]](#)
- **Winner, Health Track** — AI Crusade 2023 (First AI Hackathon of Nepal). [\[Certificate\]](#)
- **Scholarships:** Mahatma Gandhi Scholarship (Indian Embassy, 2019–2021); Merit-based scholarship, Kathmandu University (1st Semester, 2022).
- **Internal Relations & Operations Lead** — KU HackFest 2024: coordinated logistics, judging, and volunteer management for 48-hour hackathon (~200 team applications).

REFERENCES

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